

Identification of type 2 diabetes- and obesity-associated human β -cells using deep transfer learning

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eLife assessment

This is a **useful** study that applies deep transfer learning to assign patient-level disease attributes to single cells of T2D and non-diabetic patients, including obese patients. This analysis identified a single cluster of T2D-associated β -cells; and two subpopulations of obese- β -cells derived from either non-diabetic or T2D donors. The findings were validated at the protein level using immunohistochemistry on islets derived from non-diabetic and T2D organ donors, contributing **solid** experimental evidence for the computational analyses.

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Abstract

Diabetes affects >10% of adults worldwide and is caused by impaired production or response to insulin, resulting in chronic hyperglycemia. Pancreatic islet β -cells are the sole source of endogenous insulin and our understanding of β -cell dysfunction and death in type 2 diabetes (T2D) is incomplete. Single-cell RNA-seq data supports heterogeneity as an important factor in β -cell function and survival. However, it is difficult to identify which β -cell phenotypes are critical for T2D etiology and progression. Our goal was to prioritize specific disease-related β -cell subpopulations to better understand T2D pathogenesis and identify relevant genes for targeted therapeutics. To address this, we applied a deep transfer learning tool, DEGAS, which maps disease associations onto single-cell RNA-seq data from bulk expression data. Independent runs of DEGAS using T2D or obesity status identified distinct β -cell subpopulations. A singular cluster of T2D-associated β -cells was identified; however, β -cells with high obese-DEGAS scores contained two subpopulations derived largely from either non-diabetic or T2D donors. The obesity-associated non-diabetic cells were enriched for translation and unfolded protein response genes compared to T2D cells. We selected CDKN1C and DLK1 for validation by immunostaining in human pancreas sections from healthy and T2D donors. Both CDKN1C and DLK1 were heterogeneously expressed among β -cells. CDKN1C

was increased in β -cells from T2D donors, in agreement with the DEGAS predictions, while DLK1 appeared depleted from T2D islets of some donors. In conclusion, DEGAS has the potential to advance our holistic understanding of the β -cell transcriptomic phenotypes, including features that distinguish β -cells in obese non-diabetic or lean T2D states. Future work will expand this approach to additional human islet omics datasets to reveal the complex multicellular interactions driving T2D.

Background

The advent of high-throughput single-cell RNA sequencing (scRNA-seq) has enabled the generation of an array of single-cell atlases from pancreatic islets. These findings have expanded our understanding of the major cell types of the pancreas along with how they are implicated in both type 1 diabetes (T1D) and type 2 diabetes (T2D). Notably, these studies have: 1) identified multiple reliable transcriptomic markers for endocrine and exocrine pancreatic cell types; 2) provided insight into novel subtypes of cells; and 3) generated large cellular atlases spurring innovation in the development of single cell methods and analysis. scRNA-seq analysis has enabled a more robust characterization of islet cell heterogeneity which may underlie diversity in diabetes risk and drug response ^{1–5}. In published comparisons between the islet single-cell transcriptomes of humans versus mice and pigs, differences were observed in the relative proportion of major cell types as well as in many cell type-specific genes ⁶. These findings further support a focus on integrating human islet transcriptomic data to delineate disease processes and identify therapeutic targets and biomarkers.

As bulk RNA-seq and scRNA-seq human islet datasets have become increasingly available, so too has the need for new computational tools to integrate the transcriptomics and donor metadata. Some sets of islet scRNA-seq data have been combined and made searchable through web portals to browse datasets and compare cell types and marker genes ^{6–9}. To further utilize these types of integrated transcriptomic data and donor metadata, we previously developed DEGAS as a flexible deep transfer learning framework that can be used to overlay disease status, survival hazard, drug response, and other clinical information directly onto single cells. Machine learning tools like DEGAS have been primarily used in cancer datasets, but not in human pancreatic islets until now.

The reasons why some obese individuals succumb to T2D while other do not likely involves both genetic and environmental factors, but the factors that underlie this transition are incompletely understood ¹⁰. Analyses of human genomics and islet transcriptomics indicate many β -cell genes have causal roles in T2D ^{11–15}. Subsets of β -cells which are more resilient or susceptible to failure under the secretory pressure of insulin resistance may be uncovered by combining transcriptomics with machine learning approaches like DEGAS. Here we have implemented DEGAS to predict T2D- and obesity-associated subclusters of human pancreatic islet β -cells using a combination of publicly available scRNA-seq and bulk RNA-seq human islet data and associated donor metadata. Through this analysis we sought to identify genes implicated in T2D and obesity which were up- or down-regulated in subpopulations of β -cells identified by DEGAS and to validate our findings at the protein level using immunohistochemistry of pancreas tissue from non-diabetic and T2D organ donors. Our findings applying DEGAS to islet data have implications for β -cell heterogeneity in T2D and obesity.

Methods

Human islet bulk transcriptomic dataset acquisition, processing, and analysis

In this study, human islet bulk RNA-seq raw count data (aligned to GRCh38) and donor metadata from Marselli, et al. GSE159984 was downloaded from the Gene Expression Omnibus¹⁶ (Table S1). The rationale for using this dataset was its large sample size of both non-diabetic and T2D samples and the agreement of differentially-expressed genes between the GSE159984 dataset and a similarly large independent dataset¹⁷. All genes in the read count data were filtered based on the one-to-one identifier correspondences between gene symbol, Entrez gene identifier, and Ensembl identifier from the Matched Annotation from NCBI and EBI table (MANE GRCh38 v1.1)¹⁸.

Sample data were labeled and grouped by their RRID and disease status (ND, non-diabetes vs. T2D, type 2 diabetes). After filtering, the GSE159984 read count table contained 19,058 genes from N = 27 T2D samples and N = 58 non-diabetes samples (**Fig 1A**). Differential gene expression analysis between ND and T2D groups was performed using the edgeR likelihood ratio test and cutoffs were FDR < 0.05 and $|\log_2 \text{fold-change}| \geq 0.58$. The processed and filtered read count table and edgeR results are provided in Table S2 and the R script used for processing is available on GitHub (https://github.com/kalwatlab/Islet_DEGAS_v1). Selected individual genes are shown in **Fig 1H** as box plots to illustrate the quartiles across all donor samples.

Human islet single-cell data acquisition, filtering, and integration

We obtained read count tables for five single-cell human islet datasets (GSE84133; GSE85241; E-MTAB-5061; GSE81608; GSE86469 (Table S1)) which were realigned to GRCh38.p5 (hg38)⁸ (**Fig 1B**). Metadata were downloaded from GEO or obtained from the supplemental information of the respective publications. These five datasets were previously integrated and analyzed⁸. To prepare the data for our machine learning analyses, we repeated the integration and analysis of these datasets in R using Seurat to exclude cells with low expressed genes and cells with over 20% mitochondrial gene expression (low viability). The upper and lower limits varied between the datasets to account for differences in library preparation and sequencing platform following guidelines in the Seurat documentation.

Datasets were integrated into a single dataset using Seurat version 4.1.3, the Seurat objects were normalized using regularized negative binomial regression (SCTransform) to correct for batch effect and clustered using default settings. The most variable features were used to identify integration anchors which are passed to the *IntegrateData* function to return a Seurat object containing an integrated expression matrix of all cells.

The clusters were visualized using UMAP and expression of pancreatic hormone genes INS, GCG, SST, PPY, GHRL were used to annotate β , α , delta, γ and ϵ cell clusters, respectively. Clustering analysis used the Louvain algorithm¹⁹ and labelling was performed to identify different cell types because our integrated dataset contained a mixture of cells from pancreatic islets. In total, 22 clusters were identified from 17,273 pancreatic cells and the clusters were classified into endocrine and non-endocrine cell types based on cell-specific marker expression (**Fig 2A**). R scripts are available on GitHub (https://github.com/kalwatlab/Islet_DEGAS_v1). The Seurat object containing all read count matrices and merged data and metadata is available on Mendeley²⁰.

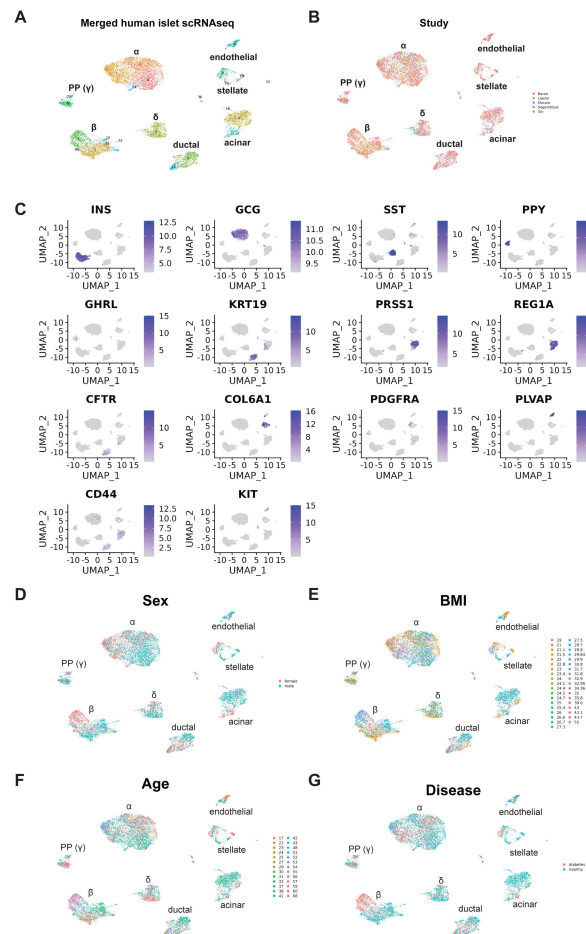


Figure 2.

Validation of merged scRNA-seq datasets from five human islet studies.

A) Seurat clustering of merged human islet scRNA-seq data from [Fig 1B](#). **B)** Study identifiers overlaid on single cell plot. **C)** Gene expression plots for the major endocrine and exocrine pancreas genes to demonstrate successful clustering of specific cell types. Metadata was overlaid onto single cell UMAP plots for sex (**D**), BMI (**E**), age (**F**), and diabetes status (**G**).

Transfer learning using DEGAS on human islet transcriptomic data

The integrated single cell and bulk datasets were processed with the DEGAS (v1.0) pipeline (<https://github.com/tsteelejohnson91/DEGAS>)²¹ to calculate disease risk scores associated with T2D status or BMI status. The bulk expression data were scaled and normalized prior to DEGAS analysis using the *preprocessCounts* function. The merged scRNA-seq expression matrix, the bulk expression matrix, and donor sample labels (matched with the bulk samples) were used as input. The intersection of highly variable genes between scRNA-seq and bulk expression data were used for further analysis. The DEGAS model was trained and predicted on the formatted data. Updated scripts and instructions for running DEGAS are available on GitHub (version 1.0) (<https://github.com/tsteelejohnson91/DEGAS>). DEGAS for T2D and BMI were run independently.

In DEGAS for T2D, the donor labels were “normal” vs “T2D.” In DEGAS for BMI metadata, donors from bulk RNA-seq were categorized and labeled as “lean” (<25 BMI), “overweight” (25-30 BMI) and “obese” (>30 BMI) according to CDC guidelines. The models were trained to calculate T2D-association scores and BMI-association scores for each of the respective categories. We identified differentially-expressed genes using *FindMarkers* function in β -cells with high vs low T2D-association scores ($\beta^{\text{T2D-DEGAS}}$) or high obesity scores ($\beta^{\text{obese-DEGAS}}$) among healthy or T2D donor single-cells.

Subclustering of β -cells and post-DEGAS scRNA-seq analysis

We isolated the β -cell cluster using *subset* function and reclustered them separately from the other islet cell types. The β -cells were further classified by different thresholding parameters including median and quantiles. For quantile and median thresholds, we generated high, medium and low, as well upper 50% and lower 50% of median. We used the *FindMarkers* function to identify differentially-expressed genes in high vs. low (Fig 4D), high vs. medium+low, and median (upper 50% vs lower 50%) (Table S3).

Pathway analysis

To analyze differentially-expressed genes that were either up- or down-regulated between clusters, enrichment analyses and plot generation were completed using ClusterProfiler, EnhancedVolcano, and ggplot2 R packages in R version 4.2.2 and RStudio. Enrichment of gene sets within the the Biological Process

GO terms are shown and cluster profiler outputs are provided in Table S4. Venn diagrams for comparing DEGAS and RePACT hit genes (Table S5) were created using <https://molbiotools.com/listcompare.php>^{22, 23}. Gene set enrichment analysis was run using GSEA software^{22, 23} downloaded from www.gsea-msigdb.org/gsea and the MSigDB Hallmark Gene Set²⁴ from the Broad Institute. For bulk RNA-seq analysis, the edgeR results containing all expressed genes was used as input to run standard GSEA. For GSEA analysis of comparisons between subpopulations of β -cells from scRNA-seq data, a ranked list based on log2fold-change was generated without a cutoff for adjusted P value. GSEAPreranked mode used the ‘classic’ enrichment statistic for 1000 permutations.

Human pancreas tissue staining and microscopy

Formalin-fixed paraffin-embedded (FFPE) de-identified human pancreas tissue were obtained through the Integrated Islet Distribution Program (IIDP) and the National Disease Research Interchange (NDRI) (Table S6). IIDP provided 5 μ m tissue sections on slides and NDRI tissue blocks were processed into 5 μ m sections and mounted on glass slides by the Indiana University School of Medicine Histology Lab Service Core. Slides were deparaffinized by xylene and ethanol washes. Antigen retrieval was performed by heating for 40 min in an Epitope Retrieval Steamer with slides

submerged in Epitope Retrieval Solution (IHC-Tek). Subsequently, slides were placed onto disposable immunostaining coverplates and inserted into the Sequenza slide rack (Ted Pella / EMS) for washing, blocking, and antibody incubations. After three 10 min washes in IHC wash buffer (0.1% Triton X-100 and 0.01% sodium azide in PBS pH 7.4), slides were blocked for 1 h at room temperature in normal donkey serum (NDS) block solution (5% donkey serum, 1% bovine serum albumin, 0.3% Triton X-100, 0.05% Tween-20, and 0.05% sodium azide in PBS pH 7.4). Slides were then incubated overnight at 4°C with primary antibodies diluted in NDS block solution (Table S7). After three washes in IHC wash buffer 200 μ L each), slides were incubated in secondary antibodies in NDS block solution for 1 h at room temperature, followed by DAPI staining for 10 min and three additional washes. The washed slides were mounted in polyvinyl alcohol (PVA) mounting medium (5% PVA, 10% glycerol, 50mM Tris pH 9.0, 0.01% sodium azide) or Prolong Gold and imaged on Zeiss LSM710 confocal microscope equipped with a Plan-Apochromat 20x/0.8 objective (#420650-9901). Images were processed in the Zeiss Zen software to add scale bars, set coloration for channels, and generate merged images. Scale bars indicate 50 μ m. Distribution of DLK1 intensity within INS⁺ areas and CDKN1C⁺ β -cell nuclei were quantified using CellProfiler (v4.2.7). Data were processed in R and plotted in GraphPad prism.

Statistical Analysis

In edgeR bulk RNAseq differential gene expression analysis, the likelihood ratio test was applied. In the FindMarkers function in Seurat the default Wilcoxon Rank Sum test was used for differential gene expression analysis. Cut-offs for significant differentially expressed genes were set at $|\log_2\text{fold-change}| > 0.58$ and adjusted p-value < 0.05 .

Results

Differentially expressed genes from T2D bulk RNA-seq human islets show inflammatory signature and correlate with independent islet datasets

We selected bulk RNA-seq data from Marselli, et al. 2020 (GSE159984) which contains 58 non-diabetic and 27 T2D samples with associated metadata for BMI, age, and sex ¹⁶ (Fig 1A-C). To determine the suitability of this data for DEGAS, we reanalyzed the read count matrix from GSE159984 using the edgeR likelihood ratio test to identify up- and down-regulated genes, visualized in both volcano (Fig 1D) and MD plot (Fig 1E) formats to highlight fold-change versus significance and versus overall expression levels, respectively (Table S2). We observed altered expression of genes well-known to be dysregulated in T2D, including *IAPP* ^{25, 26}, *PAX4* ^{27, 28}, *SLC2A2* ²⁹, *FFAR4* ³⁰, and *ENTPD3* ^{31, 32} (Fig 1D-E).

Gene set enrichment analysis (GSEA) indicated that the gene expression profile of T2D human islets from this dataset was associated with an inflammatory response phenotype (Fig 1F). To compare our analysis of Marselli, et al. with an independent cohort, we selected a bulk RNA-seq human islet transcriptomic analysis from Asplund et al. ¹⁷. We compared the published supplementary table of log₂-fold changes for differentially expressed genes and to the Marselli dataset (Fig 1G) (Table S2). Significant differentially expressed genes from both Marselli and Asplund data included up-regulation of *SFRP4* and *PODN* (Fig S1A-B), and down-regulation of *UNC5D* and *FFAR4* (Fig S1C-D). We also found substantial agreement in the directional changes of most other differentially expressed genes from both datasets, indicating that the Marselli islet data represents a suitable cohort for DEGAS analysis. Finally, we compared up- and down-regulated genes from T2D islets in the Marselli dataset with T2D effector genes from the T2D Knowledge Portal ³³. Down-regulated genes *PAX4* and *SLC2A2* and up-regulated genes *APOE*

each have evidence of a strong or causal role in T2D at the genetic level, while *CYTIP* was up-regulated in T2D islets and has intron variant SNPs associated with T2D (e.g. rs13384965) ³⁴ (Fig 1H).

DEGAS revealed T2D- and obesity-associated β -cells and marker genes within integrated human islet scRNA-seq data

We obtained five realigned scRNA-seq datasets ⁸ including Baron (GSE84133), Muraro (GSE85241), Segerstolpe (E-MTAB-5061), Lawlor (GSE86469), and Xin (GSE81608), and integrated them using Seurat 4.3.0 to subtype each of the major cell types for use with DEGAS. All five datasets were successfully integrated resulting in 17,273 cells (Fig 2A-B). Different cell types were clearly stratified from the scRNA-seq data, as indicated by distinct clusters of cell types enriched for their respective markers including β -cells (*INS*), α -cells (*GCG*), δ -cells (*SST*), ϵ -cells (*GHRL*), PP-cells (*PPY*), acinar cells (*REG1A*, *PRSS1*), ductal cells (*KRT19*, *CFTR*), stellate cells (*COL6A1*), endothelial cells (*PLVAP*), and mesenchymal cells (*CD44*) (Fig 2C). Other parameters including sex, BMI, age, and T2D status were represented in the merged dataset (Fig 2D-G). Markers of T2D vs. ND for all cells and for only β -cells were also extracted for downstream comparisons (Fig S2A-B). There was limited overlap between the differentially expressed genes in Marselli, et al. compared to T2D vs. ND for all single cells, but included *IAPP*, *ENTPD3* and *FFAR4*.

Next, we implemented DEGAS ²¹, a unique tool that leverages the increased sequencing depth, larger number of clinical covariates and sample sizes of bulk sequencing data. We used the deep learning architecture within DEGAS to project bulk islet sequencing data into the same latent representation as single cell islet data via domain adaptation. From this common latent representation, clinical covariates from the bulk sequencing data (e.g. T2D status and BMI) were projected onto single cells in a process called transfer learning. This process generates unitless disease-association scores for each single cell. In parallel, we applied DEGAS to the merged scRNAseq data using either T2D or obesity as the scoring feature.

T2D-DEGAS

First, we next mapped the T2D-association scores onto the single cells (Fig 3A). β -cells had a wide distribution of scores, possibly reflecting β -cell heterogeneity or altered β -cell gene expression after onset of T2D (Fig 3B). We focused on differential T2D-association scores only among β -cells by subsetting them based on *INS* expression and superimposed the T2D-association scores onto each β -cell in the UMAP plot (Fig 4A). β -cells with higher scores (pink coloration) transcriptionally associate with human islet T2D expression profiles more than the lower scoring (black) β -cells. Next, we explored median and quantile thresholds for the β -cell T2D-association scores (Fig S3A-B) and continued our analysis with the cells grouped into high (upper 20%), medium and low (lower 20%) categories (Fig 4B). We compared the high versus low $\beta^{\text{T2D-DEGAS}}$ groups to genes enriched in either subset of β -cells (Fig 4C, Table S3). High-scoring $\beta^{\text{T2D-DEGAS}}$ cells had elevated expression of *CDKN1C* (p57^{Kip2}) which binds and inhibits G1 cyclin/CDK complexes and is also involved in cellular senescence ^{36, 37}, and reduced expression of *IAPP*, consistent with results from bulk T2D islet expression data (Fig 1G). These findings show that the DEGAS can successfully identify diabetes-relevant genes enriched in specific subclusters.

As an additional way to highlight genes that were identified by DEGAS, we excluded genes that were differentially expressed between all non-diabetic (ND) and T2D β -cells (based on the merged scRNA-seq donor metadata Fig S4D, Table S3). This process left over 90% of significant differentially expressed genes remaining which were only identified through DEGAS analysis (Fig 4D). This emphasized the advantage of using DEGAS to generate continuous variables (disease-association scores) which enable thresholding and prioritizing cellular subpopulations. Gene

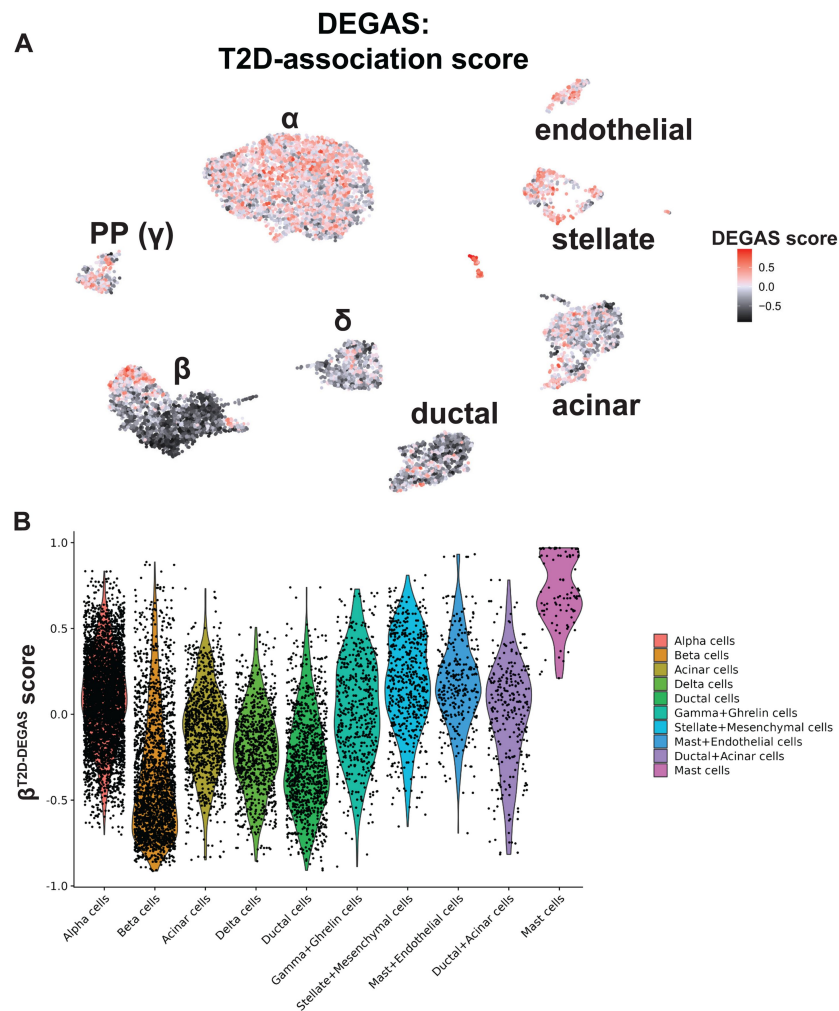


Figure 3.

DEGAS analysis based on T2D status.

A) DEGAS T2D association scores (T2D-DEGAS) for each cell was overlaid onto the single cell UMAP plot. Higher scores in pink/red indicate a strong positive association and negative scores in darker black indicate a negative association of those cells with T2D. **B)** Violin plot displaying the aggregate T2D-DEGAS scores per cell type.

ontology (GO) analysis on high-scoring $\beta^{\text{T2D-DEGAS}}$ cells showed enrichment in biological process categories including neutrophil degranulation, cell growth, and negative regulation of cell development (**Fig 4E** [↗](#), Table S4). These categories were driven in part by immune-related genes (e.g. *HLA-B*, *PSAP*) and cell cycle regulator *BTG1*. Conversely, GO analysis on low-scoring $\beta^{\text{T2D-DEGAS}}$ cells showed enrichment of categories primarily related membrane protein targeting, translation, and mitochondrial electron transport (**Fig 4F** [↗](#), Table S4). The top six enriched GO categories were driven largely by *RPS/RPL* genes. We also performed ranked-list GSEA on these same high- vs. low-scoring $\beta^{\text{T2D-DEGAS}}$ subpopulations which largely agreed with the GO analysis (**Fig 4G** [↗](#)). We observed significant positive enrichment for hypoxia, TNF α signaling, estrogen response, and glycolysis (**Fig S3C** [↗](#)), while significant negative enriched pathways included overlapping genes in oxidative phosphorylation and Myc-target pathways (**Fig S3D** [↗](#)). Examples of genes that drive the GO and GSEA categories are shown (**Fig 4H** [↗](#)) and heterogeneity in the expression in a selection of these genes can be seen in β -cell UMAP plots (**Fig S3E** [↗](#)).

Obese-DEGAS

Second, we applied DEGAS using obesity as the disease feature. Obesity and T2D are related, but not mutually inclusive. We surmised that obesity-association scores in β -cells may highlight unique subpopulations compared to the $\beta^{\text{T2D-DEGAS}}$ cells. Therefore, we categorized bulk RNA-seq donors by BMI (lean, <25; overweight, 25-30; obese >30) and ran the DEGAS analysis based on these categories to calculate BMI-association scores similar to our approach with T2D. Overlaying the BMI-DEGAS scores onto the islet single-cell data showed a distinct pattern of cell labeling between **Fig S4A-B** [↗](#)). The cell types with the highest obesity-association scores were β -cells, acinar cells, and stellate cells (**Fig 5B** [↗](#)).

We next reclustered β -cells with overlaid lean, overweight, and obesity-association scores (**Fig S4C-D** [↗](#), **6A** [↗](#)). It is notable that the subpopulation of $\beta^{\text{lean-DEGAS}}$ cells appear to overlap with the subpopulation of low-scoring $\beta^{\text{T2D-DEGAS}}$ cells ([↗](#)), suggesting that single β -cells share features of low BMI and potentially lower T2D association. $\beta^{\text{overweight-DEGAS}}$ scores did not highlight any subpopulation of β -cells (**Fig S4D** [↗](#)). However, we observed two subpopulations of high-scoring $\beta^{\text{obese-DEGAS}}$ cells (**Fig 6A** [↗](#)), a feature that did not occur with $\beta^{\text{T2D-DEGAS}}$ cells (**Fig 4B** [↗](#)). We observed that a distinguishing difference between these two groups of obesity-associated β -cells was the donor diabetes status (ND vs T2D) from within the single-cell metadata. We applied a median quantile threshold to extract these two subpopulations (**Fig S4E** [↗](#)) and classified the cell clusters based on their donor diabetes status (**Fig 6B** [↗](#), **Fig S4F** [↗](#)) to create ND- $\beta^{\text{obese-DEGAS}}$ and T2D- $\beta^{\text{obese-DEGAS}}$ subpopulations (**Fig 6C** [↗](#)). We next determined differentially expressed genes between the ND- $\beta^{\text{obese-DEGAS}}$ and T2D- $\beta^{\text{obese-DEGAS}}$ cells (**Fig 6D** [↗](#)). Gene ontology analysis of genes upregulated in the ND- $\beta^{\text{obese-DEGAS}}$ cluster indicated an enrichment for unfolded protein response (UPR) processes like vesicle transport, translation, and protein folding and stability (**Fig 6E** [↗](#)). Specifically, the ND- $\beta^{\text{obese-DEGAS}}$ cluster was enriched for expression of adaptive (e.g. *MANF* and *HSPA5*) and maladaptive (e.g. *TRIB3*, *DDIT3*) UPR genes, as well as genes involved in ER-associated degradation (e.g. *SEL1L*, *DERL2*, and *SDF2L1*). Gene ontology analysis of genes upregulated in the T2D- $\beta^{\text{obese-DEGAS}}$ cluster showed enrichment for hormone transport and secretion pathways (e.g. *IGFBP5*, *UCN3*, *G6PC2*) and inflammatory/immune-related pathways (e.g., *HLA-A/B/E*, *STAT3*) (**Fig 6F** [↗](#)). In parallel, we performed ranked-list GSEA on the ND- and T2D- $\beta^{\text{obese-DEGAS}}$ subpopulations which was largely consistent with gene ontology results (**Fig 6G** [↗](#)). Significantly enriched pathways for ND- $\beta^{\text{obese-DEGAS}}$ included the UPR and proliferation (E2F and MTORC signaling hallmarks) (**Fig S4G** [↗](#)), while T2D- $\beta^{\text{obese-DEGAS}}$ enriched pathways were similar to that of high-scoring $\beta^{\text{T2D-DEGAS}}$ cells and included hypoxia and glycolysis (**Fig S4H** [↗](#)). Examples of genes driving the GO and GSEA categories are shown, as well as *CDKN1C* and *DLK1*, given their differential enrichment in both $\beta^{\text{T2D-DEGAS}}$ and T2D- $\beta^{\text{obese-DEGAS}}$ cells (**Fig 6H** [↗](#)).

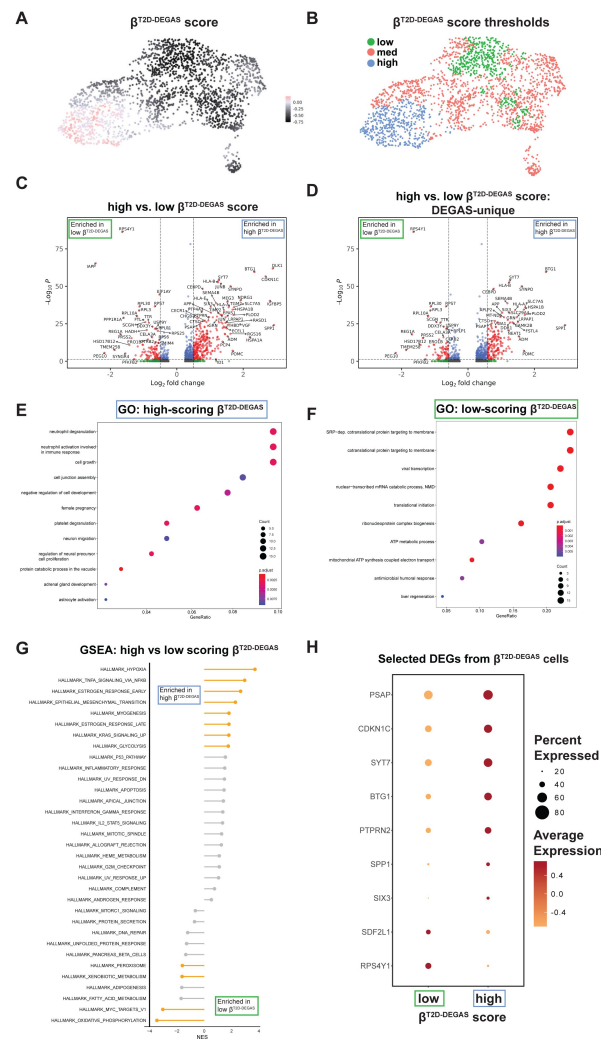


Figure 4.

Identification of differentially-expressed genes in high- and low-scoring $\beta^{T2D-DEGAS}$ cell populations.

A) DEGAS scores for T2D association ($\beta^{T2D-DEGAS}$) were overlaid onto the cell plot. **B)** β -cells were classified as low, medium and high-scoring $\beta^{T2D-DEGAS}$ subpopulations for downstream analysis. **C)** Differential expression analysis of high vs. low $\beta^{T2D-DEGAS}$ scores. Genes with $p < 0.05$ and > 1.5 -fold change are highlighted in red. **D)** Genes from (C) were filtered to remove DEGs that could be identified by comparing β -cells of T2D vs non-diabetic donors in the single cell data (**Fig S2B**). Gene ontology (GO) analysis of genes enriched in the high (**E**) and low (**F**) scoring $\beta^{T2D-DEGAS}$ subpopulations. **G)** GSEA results for high- vs. low-scoring $\beta^{T2D-DEGAS}$ subpopulations. **H)** Bubble plot highlighting genes driving GO and GSEA categories.

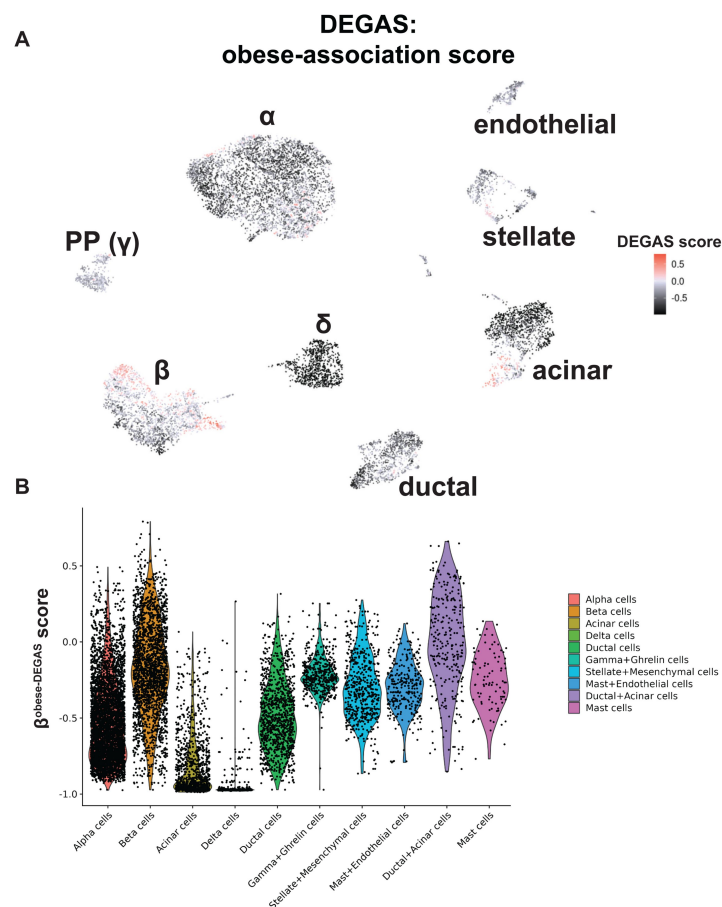


Figure 5.

DEGAS analysis based on obesity status.

A) DEGAS obesity association scores (obese-DEGAS) for each cell was overlaid onto the single cell UMAP plot. Higher scores in pink/red indicate a strong positive association and negative scores in darker black indicate a negative association of those cells with obesity. **B)** Violin plot displaying the aggregate obese -DEGAS scores per cell type.

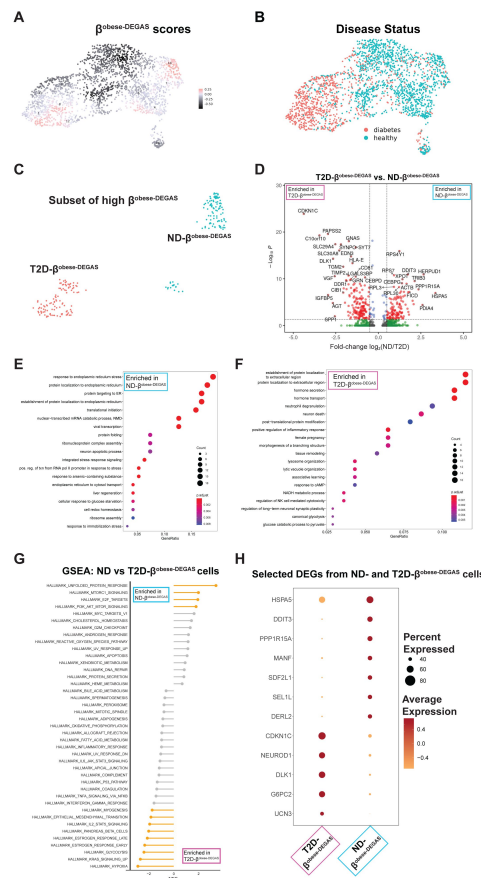


Figure 6.

Two subpopulations of high $\beta^{\text{obese-DEGAS}}$ cells defined by underlying single cell donor T2D status uncovers stress signature in non-diabetic high-scoring obese-DEGAS β -cells.

A) β -cells were subsetted from [Fig 5A](#) based on *INS* expression and obese-DEGAS scores overlaid onto the single cells. **B)** Donor diabetes status overlaid onto the β -cells. **C)** β -cells were further subsetted based on both their high obese-DEGAS scores and by the underlying T2D status of the single cell donors, leading to two major subpopulations, ND- $\beta^{\text{obese-DEGAS}}$ and T2D- $\beta^{\text{obese-DEGAS}}$ cells. **D)** Differential expression analysis comparing ND- $\beta^{\text{obese-DEGAS}}$ and T2D- $\beta^{\text{obese-DEGAS}}$ subpopulations. Genes with $p < 0.05$ and > 1.5 -fold change are highlighted in red. Gene ontology (GO) analysis of genes enriched in the ND- $\beta^{\text{obese-DEGAS}}$ (**E**) and T2D- $\beta^{\text{obese-DEGAS}}$ (**F**) subpopulations. **G)** Gene set enrichment analysis (GSEA) of ND- vs. T2D- $\beta^{\text{obese-DEGAS}}$ subpopulations. **H)** Bubble plot highlighting genes driving GO and GSEA categories.

Combined, our results applying DEGAS using T2D and obesity as disease features support the idea that the identified genes and pathways may underlie heterogeneity in β -cell resilience or susceptibility to obesity-related stress in the context of T2D development. In particular, *DLK1* and *CDKN1C* ranked highly in both T2D-DEGAS and obese-DEGAS analyses (**Fig S4I** [↗](#)), making them reasonable candidates for validation of this approach.

Heterogeneous expression of *DLK1* and *CDKN1C* in human pancreatic islets from diabetic and non-diabetic donors with lean, overweight, or obese BMI

To support the DEGAS approach for identifying true markers of β -cell subpopulations or heterogeneity associated with diabetes, we selected *DLK1* and *CDKN1C* as candidate genes for immunostaining in FFPE human pancreas sections from non-diabetic and T2D donors. We chose Delta-Like 1 Homolog (*DLK1*) because of its strong associations with T1D [38](#) [↗](#) and T2D [39](#) [↗](#). *DLK1* expression overlapped with high scoring $\beta^{\text{T2D-DEGAS}}$ cells (**Fig 7A** [↗](#)) and with T2D- $\beta^{\text{obese-DEGAS}}$ cells (**Fig 7B** [↗](#)).

DLK1 immunostaining primarily colocalized with β -cells in non-diabetic human pancreas and exhibited heterogeneity in its expression intensity (**Fig 7C** [↗](#), **S5** [↗](#)). Some islets had *DLK1*/INS co-staining in many β -cells while others only a few *DLK1*⁺ β -cells and this pattern did not appear to be associated with BMI. In T2D pancreas, *DLK1* expression appeared to be reduced in some donors, however an analysis of the distribution of *DLK1* staining intensity across INS⁺ cells indicated that there was no major difference between ND and T2D (**Fig 7D** [↗](#)). We chose our second candidate, Cyclin-Dependent Kinase inhibitor 1C (*CDKN1C*), because it has been linked to β -cell replication [40](#) [↗](#). *CDKN1C* expression also overlapped with high scoring $\beta^{\text{T2D-DEGAS}}$ cells (**Fig 7E** [↗](#)) and with T2D- $\beta^{\text{obese-DEGAS}}$ cells (**Fig 7F** [↗](#)). *CDKN1C* staining was primarily nuclear and was heterogeneous across β -cells in non-diabetic human pancreas (**Fig 7G** [↗](#)). In agreement with the DEGAS results, *CDKN1C* staining was increased in β -cell nuclei from T2D donors as quantified by CellProfiler analysis (**Fig 7H** [↗](#)). BMI status did not appear to correlate with *CDKN1C* expression in either non-diabetic or T2D donors.

Discussion

Deep transfer learning is a useful approach to identify disease-associated genes in subsets of heterogeneous islet cells

The rapid advancement of deep learning has unveiled new opportunities in diabetes research, specifically the ability to merge and analyze the large amount of publicly available omics data. The DEGAS deep transfer learning framework provides an extremely powerful and versatile tool allowing for time to event outcomes and classification outcomes to be transferred to individual cells. Notably, this direct transfer to single cells avoids the cluster resolution problems that arise using deconvolution approaches, allowing for subsets of cells within a cluster to be assigned different associations with clinical outcomes. In this way, DEGAS allows the definition of essentially new cell subpopulations that would not necessarily be identified by standard clustering algorithms and these subpopulations are rationally selected based on assigned disease association scores. A key point from our study is that the disease-associated single cells identified by DEGAS are those cells whose transcriptomic signature share complex patterns with patient transcriptomics that have known clinical and metabolic attributes. Because of this, the genes identified as up- or down-regulated in subsets of β -cells may be: 1) altered independent of disease due to genetic background of donors; 2) altered downstream of the obese or diabetic states; or 3)

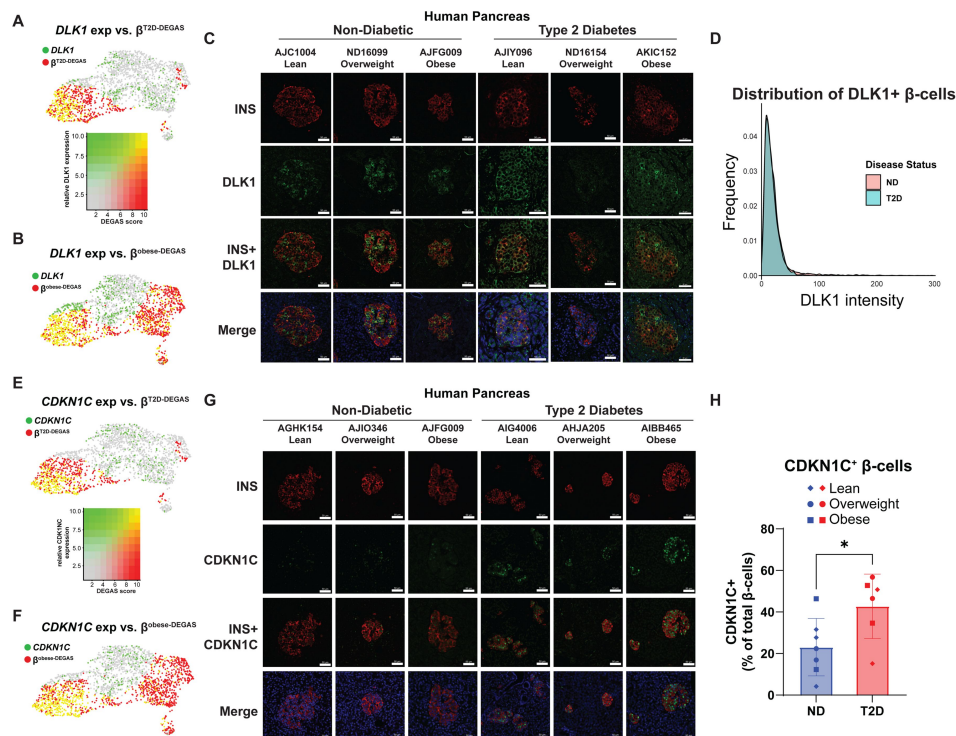


Figure 7.

DLK1 is a β -cell gene heterogeneously expressed between cells and islets of non-diabetic and T2D humans.

A) DLK1 expression and $\beta^{T2D-DEGAS}$ scores overlaid onto the β -cell cluster. Yellow dots indicate cells with high DLK1 expression and high $\beta^{T2D-DEGAS}$ score, as shown in the adjacent overlay heatmap. **B)** DLK1 expression and $\beta^{obese-DEGAS}$ scores overlaid onto the β -cell cluster. Yellow dots indicate cells with high DLK1 expression and high $\beta^{obese-DEGAS}$ score. **C)** Formalin-fixed paraffin-embedded (FFPE) sections of human pancreas from non-diabetic (ND) and T2D donors were stained for INS and DLK1. Representative images are shown from three different ND and T2D donors. **D)** CellProfiler analysis showing distribution of DLK1 in ND vs T2D donors. UMAP plot overlaying CDKN1C expression and $\beta^{T2D-DEGAS}$ (**E**) or $\beta^{obese-DEGAS}$ (**F**) scores. **G)** Formalin-fixed paraffin-embedded (FFPE) sections of human pancreas from non-diabetic (ND) and T2D donors were stained for INS and CDKN1C. Representative images are shown from three different ND and T2D donors. **H)** Increased CDKN1C expression observed in β -cells from T2D patients. *, $P < 0.05$ by Student's T-test.

representative of resilient or dysfunctional β -cells in the face of metabolic syndrome. Through careful cohort inclusion criteria and batch effect removal, our analyses aim to limit these caveats when possible.

Various deconvolution methods enable estimating relative proportions of cell types in bulk samples using scRNA-seq data (e.g. BSEQ-sc ¹, MuSiC ⁴¹, CIBERSORT ⁴², and DWLS ⁴³). These methods have been instrumental in identifying changes to relative quantity of cell types within normal and disease tissues; however, they do not provide detailed information about subsets of cell types which may be related to disease and are limited to the resolution of the predefined cell types through clustering. Early attempts to address this include unsupervised learning algorithms (e.g. RaceID) ⁴⁴. Now, there are multiple advanced computational approaches to identify subsets of cells related to disease status, survival, drug response, and other disease metrics. These tools can be broadly categorized as ‘cell prioritization algorithms’ and include: DEGAS ²¹, scAB ⁴⁵, Scissor ⁴⁶, and scDEAL ⁴⁷. Scissor and scAB can assign survival or clinical information from bulk expression data to disparate scRNA-seq datasets using regression and matrix factorization, respectively. scDEAL is a deep learning framework specifically designed for the assignment of drug response information from bulk expression data to subsets of single cells. The major advantages of DEGAS are 1) the ability to select subsets of β -cells at an individual cell resolution based on their disease association; 2) unique qualities of the highly non-linear DEGAS technology vs. linear models of other tools to detect complex molecular programs; and 3) speed and scalability based on efficient implementation. DEGAS has the potential to be used on datasets containing well over 1 million cells, unlike other methods.

Nevertheless, certain linear model-based approaches, like RePACT (regressing principal components for the assembly of continuous trajectory), have successfully uncovered genes associated with β -cell heterogeneity in T2D and obesity ⁵. In that study, T2D and obesity ‘trajectory’ genes were identified. We compared the genes enriched in our high-scoring $\beta^{\text{T2D-DEGAS}}$ and in our T2D- $\beta^{\text{obese-DEGAS}}$ cells with the corresponding T2D and obesity trajectory genes from Fang et al., respectively (Fig S6, Table S5). We noted substantial agreement between DEGAS and RePACT, for example the transcription factor *SIX3* was enriched in T2D- $\beta^{\text{obese-DEGAS}}$ cells and in Fang et al.’s obese trajectory β -cells ⁵. *SIX3* has been implicated as a diabetes risk gene and is important for β -cell function ^{48, 49}. Additionally, both approaches highlighted the association of *DLK1* with obesity, but only DEGAS identified *DLK1* association with T2D. Overall, the directionality of T2D or obesity gene associations agreed between DEGAS and RePACT results. However, not all the same genes were identified by both approaches, for example, *FXYD2* was identified via RePACT, but not in DEGAS, although *FXYD2* is a down-regulated gene in T2D in Marselli et al. and Asplund et al. (Fig 1G). Interestingly, *FXYD2* has been recently shown to be a downstream effector gene of *HNF1A* in single human β -cells ⁵⁰. Down-regulation of *HNF1A* in T2D β -cells, and reduced *FXYD2* expression as a result, may contribute to membrane hyperpolarization and reduced function ⁵⁰. Distinct from the application of RePACT in Fang, et al. ⁵, we did not observe any differences in *HNF1A* expression in our DEGAS analyses and *FXYD2* was slightly enriched, but not significantly altered, in T2D- $\beta^{\text{obese-DEGAS}}$ cells. Taken together, it is necessary to apply multiple approaches to merged sets of publicly available data because each approach will likely uncover unique and important subsets of β -cells or specific genes.

Differences between high- and low-scoring $\beta^{\text{T2D-DEGAS}}$ subpopulations

High-scoring $\beta^{\text{T2D-DEGAS}}$ cells had enrichment of pathways like hypoxia and TNF α signaling which contain some overlapping genes like *BTG1*, *BTG2*, and *BHLHE40*. *BHLHE40* was recently shown to cause hypoxia-induced β -cell dysfunction via interfering with *MAFA* expression ⁵¹. *BTG1* and *BTG2* are known as anti-proliferation factors. Given the expression of both *BTG1* and *CDKN1C* in high-scoring $\beta^{\text{T2D-DEGAS}}$ cells, it is possible these cells have altered proliferative potential, even in the context of β -cells which are well-known to have low proliferation. Although some genes like

CDKN1C can be identified as differentially expressed in simple comparisons of all T2D vs ND donor β -cells in scRNA-seq data, DEGAS has the advantage of prioritization or ranking of candidates. For example, *CDKN1C* is within the top ten enriched candidates (by adjusted P value and fold-change) in the $\beta^{\text{T2D-DEGAS}}$ cluster but is 286th when simply comparing all T2D vs ND single β -cells (Table S3). Additionally, deletion of *CDKN1C* has been shown to improve human islet function⁴⁰ and gain-of-function mutants are associated with diabetes development and hyperinsulinism^{52, 53}. In agreement with our results, we found *CDKN1C* was significantly increased in β -cells from T2D donors (**Fig 7H**). From our analysis we cannot determine whether this is a response to the diabetic milieu in those donors or if *CDKN1C* is increased early and prevents β -cell compensation, further contributing to T2D pathogenesis. Understanding the differences between *CDKN1C*⁺ and *CDKN1C*⁻ human β -cells would be of interest in future work to provide insights into the regulation of β -cell replication.

The low-scoring $\beta^{\text{T2D-DEGAS}}$ cells were enriched in pathways involving oxidative phosphorylation and Myc targets. We also noted enrichment of many small and large ribosomal genes (*RPS/RPL*) in this subpopulation. *RPS/RPL* ribosomal genes are known to be highly expressed in primary human β -cells⁵⁴, and glucose stimulation induced translation of over 50 different *RPS/RPL* genes in human β -cell line EndoC- β H2⁵⁵. This suggests that low-scoring $\beta^{\text{T2D-DEGAS}}$ cells may have relatively increased translation and metabolic activity. Taken together, high-scoring $\beta^{\text{T2D-DEGAS}}$ cells may be reflective of dysfunctional β -cells in either a pre-T2D or T2D-onset state, while low-scoring $\beta^{\text{T2D-DEGAS}}$ cells could potentially represent more resilient β -cells.

Differences between ND- $\beta^{\text{obese-DEGAS}}$ and T2D- $\beta^{\text{obese-DEGAS}}$ subsets of β -cells

Our understanding of why some individuals with obesity develop T2D while others do not is incomplete. A 2021 NHANES report indicated a 41.9% prevalence for obesity in US adults aged 20 and over, with a 14.8% prevalence for diabetes⁵⁶, in line with CDC reports⁵⁷. This suggests that many individuals with obesity have either not progressed, or may never progress, to T2D. This phenomenon may be related to those individuals categorized as metabolically healthy obese as compared to metabolically unhealthy obese⁵⁸. Expanded work with DEGAS may uncover specific genes that underly this relationship. In our analysis we were able to stratify based on T2D status of the single cell donors and overlay the relative obesity-association scores from DEGAS. That analysis allowed us to set rational thresholds for grouping subsets of β -cells to compare ND- and T2D- $\beta^{\text{obese-DEGAS}}$ cells. Perhaps counterintuitively, UPR genes were highly enriched in the ND- $\beta^{\text{obese-DEGAS}}$ cells as opposed to the T2D- $\beta^{\text{obese-DEGAS}}$ cells. It is possible these ND- $\beta^{\text{obese-DEGAS}}$ cells are in a stressed pre-diabetic state on the path to T2D. Dominguez-Gutierrez et al. analyzed islet scRNA-seq data using pseudotime analysis and identified three major β -cell states which were defined by their insulin expression and UPR level⁵⁹. The authors speculate that β -cells periodically pass through these states, which in scRNA-seq are visualized as subpopulations of cells. Previously, a *FTH1* subpopulation of β -cells was identified with implications in the UPR⁶⁰. In agreement with those results, we found *FTH1* was upregulated in ND- $\beta^{\text{obese-DEGAS}}$ cells. Therefore, another possibility is that the UPR induced in ND- $\beta^{\text{obese-DEGAS}}$ cells marks a population with lower functionality but higher proliferative potential to combat the insulin resistant obese state, and the relatively lower UPR in T2D- $\beta^{\text{obese-DEGAS}}$ cells possibly correlates with enhanced functionality in established T2D. Concordantly, T2D- $\beta^{\text{obese-DEGAS}}$ cells had enrichment of hormone secretion genes (e.g. *SYT7*, *G6PC2*, *NEUROD1*, *UCN3*, *FFAR1*) in our pathway analysis (**Fig 6F**). In a study of human, mouse, and pig islet scRNA-seq data, subpopulations of ‘stressed’ β -cells were identified that exhibited enriched hallmark pathways similar to what we observe in ND- $\beta^{\text{obese-DEGAS}}$ cells⁶. Thus, multiple studies including our current analysis support the existence of subpopulations of stressed β -cells, whether transient or stable, that could in principle be targeted by therapeutics.

We also looked for enrichment of potential secreted biomarkers in high-scoring $\beta^{\text{T2D-DEGAS}}$ and $\text{T2D-}\beta^{\text{obese-DEGAS}}$ cells and identified *LRPAP1* and *C1QL1*. *LRPAP1* encodes the LDL receptor-related protein-associated protein 1 (LRPAP) and is enriched in this subpopulation. LRPAP is ubiquitously expressed and is predicted to be an ER resident protein⁶¹. LRPAP appears to be heterogeneously expressed among islet cells and was detected in human plasma in the in the Human Protein Atlas⁶². *C1QL1* encodes a secreted peptide that is highly-expressed in human islets⁶³ and was enriched in $\beta^{\text{T2D-DEGAS}}$ cells, although little else is known about its role in β -cells. Increased C1QL1 release could potentially signal through its GPCR BAI3 to suppress insulin secretion of surrounding β -cells, as has been shown for C1QL3⁶⁴.

Potential for DEGAS in identifying β -cell heterogeneity markers

Subsets or subpopulations of β -cells are an emergent property of β -cell heterogeneity. Two of the most widely used scRNA-seq datasets have also identified significant heterogeneity within pancreatic cell types, notably β -cells^{7,60}. Various markers have been identified to define these β -cell subpopulations, including expression of ST8SIA1 and CD9⁶⁵ or Flattop expression⁶⁶. Additionally, UCN3 marks mature β cells⁶⁷ and RBP4⁺ β -cells correlated with reduced function⁶⁸. Functional heterogeneity has also been described as in the case of leader/first-responder β cells^{69,70} or hub β cells⁷¹, and has been linked to transcription factors including PDX1⁷² and HNF1A⁵⁰. Nevertheless, cellular heterogeneity within a single cell type is complicated and there is uncertainty in the single cell analytics field about what constitutes a stable cell type versus a transient cell state⁷³. Tying transcriptionally distinct clusters to function or morphology may be key to making this distinction⁷⁴. By inferring disease outcomes in high resolution cellular subtypes, application of DEGAS can help to inform this debate.

In our analysis, the expression of *DLK1* and *CDKN1C* was enriched in high-scoring $\beta^{\text{T2D-DEGAS}}$ and in $\text{T2D-}\beta^{\text{obese-DEGAS}}$ cells. Recent *in vitro* work has also highlighted *DLK1* and *CDKN1C* as having increased expression in human pancreatic islets treated with 30mM glucose for 48 h⁷⁵. In agreement with scRNAseq data, we also observed heterogeneous *DLK1* and *CDKN1C* immunostaining of β -cells from human pancreas sections. *DLK1* was previously noted to be enriched in subpopulations of β -cells^{1,2,4,8,76}, however its heterogeneity in transcript and protein abundance in non-diabetic versus T2D human islet β -cells had not been described until now. *DLK1* is a maternally-imprinted gene⁷⁷ with described roles in Notch signaling^{78–80} and glucose homeostasis⁸¹. Supporting an active role β -cell disease, a recent preprint reported *DLK1* was required for proper maturation, function, and stress resilience of β -like cells differentiated from human embryonic stem cells⁸². Our results combined with the published and available preprint data suggest the role of *DLK1* in β -cell development and function may be nuanced, given that we did not observe a consistent up or down regulation at the protein level in human islets. However, *DLK1* may serve as a surface marker of, or be released from, stressed or at-risk β -cells, but increased sample sizes and in-depth analyses of human islet samples will be required to better describe these possible roles. Our DEGAS results were confirmed however with *CDKN1C* expression being increased in T2D β -cells. Other work has shown that suppression of *CDKN1C* expression in hyperglycemic, immunodeficient mice increased β -cell replication by three fold and those newly replicated cells retained mature β -cell functionality, indicating that manipulating *CDKN1C* expression can cause β -cell expansion in T2D patients⁴⁰. Additionally, loss-of-function mutations in *CDKN1C* can cause focal congenital hyperinsulinism, wherein increased β -cell proliferation and IGF-II expression are observed⁸³.

Applying tools like DEGAS to publicly available data will increase our understanding of the underlying β -cell features associated with progression to T2D or with an obese non-T2D state. Expanded DEGAS analyses will be needed to include scRNA-seq data from islets subjected to various models of T1D/T2D (e.g. cytokines, glucolipotoxicity). Major questions include whether the gene candidates identified by DEGAS are protective, disease-causing, or simply markers in β -cell subpopulations. In the future, these questions may be addressed by functional validation studies or observations in genome-wide association studies.

Limitations of the study and future directions

We think the approach of joining islet omics data to deep learning and artificial intelligence is an area in need of increased attention; however, we appreciate that our current study has limitations. First, our analysis utilized a single bulk RNA-seq human islet dataset, although the Marselli study contains a relatively large number of islet donors compared to most available datasets. Second, we merged only five scRNA-seq human islet datasets for this application of DEGAS. The datasets were chosen because of the extensive meta-analyses to which they have already been subjected⁸³, and their use as benchmarking data in many scRNA-seq analysis tools^{84–86}. scRNA-seq has a limitation on the number of detectable genes in each cell, as compared to bulk RNA-seq samples. While these limitations may eventually be mitigated by improved technologies, we can begin to overcome these by merging a larger number of human islet bulk and scRNA-seq studies which contain more donors. Increased sample sizes and study variables come with their own challenges, however the future gene candidates identified by DEGAS will be of even higher confidence. Additionally, scRNA-seq is a snapshot in time and it may be tenuous to claim that a particular cell state is transient or persistent. To support a claim that a given subset/subtype of β -cells represents an actual stable population in scRNA-seq data requires at least finding that the cells comprising the candidate population occur across multiple individual donors. In future studies, our use of the diabetes field's considerable investment in islet transcriptomics data and state-of-the-art cell prioritization tools will enable simultaneous identification of disease-associated cellular subtypes and associated biomarkers of function and dysfunction. Additionally, increasing the number of human pancreas donor tissue samples for high-content image analysis will improve confidence in candidate gene validation.

Conclusions

DEGAS has the potential to be applied to even larger mergers of single cell data (> 1 million cells) using the newly developed DEGAS atlas implementations. A vast amount of human (and other species) islet single cell transcriptomics data is publicly available, but often requires substantial reformatting of metadata and realigning of reads to be harmonized. As our endeavors and those of others proceed, DEGAS will lead to even higher confidence predictions of β -cell subpopulations. It is also apparent from our DEGAS analyses that other non- β -cells are highlighted in the islet single cell map for associations with T2D and obesity. Although outside the scope of this work, further exploration of our current DEGAS analyses, or of analyses using larger single cell integrations, will have implications for the other major islet and pancreas cell types, including α -cells, δ -cells and PP-cells. For example, PP cells have been shown to have a role in pancreatogenic diabetes as opposed to T2D⁸⁷, and there appear to be subpopulations of high and low scoring PP^{T2D}-DEGAS cells within the DEGAS analysis (**Fig 3A**). Our successful application of deep transfer learning in human islet data opens the possibility of predicting subtypes of β -cells in other diseases, like T1D and congenital hyperinsulinism, to find potential biomarkers and therapeutic candidates. In some rare diseases, like congenital hyperinsulinism, single cell data is not available, however spatial transcriptomics (ST) could be applied to FFPE sections. DEGAS has already been implemented to analyze ST data in prostate cancer and diabetic pancreas^{88–90}, and will be increasingly applicable to human pancreas samples as ST technology continues to advance. It is important to note that DEGAS is just one of many machine learning approaches to analyze transcriptomic data. It will be important to include DEGAS in combination with both linear and other non-linear models to capture as many relevant gene candidates as possible.

Supplemental Information

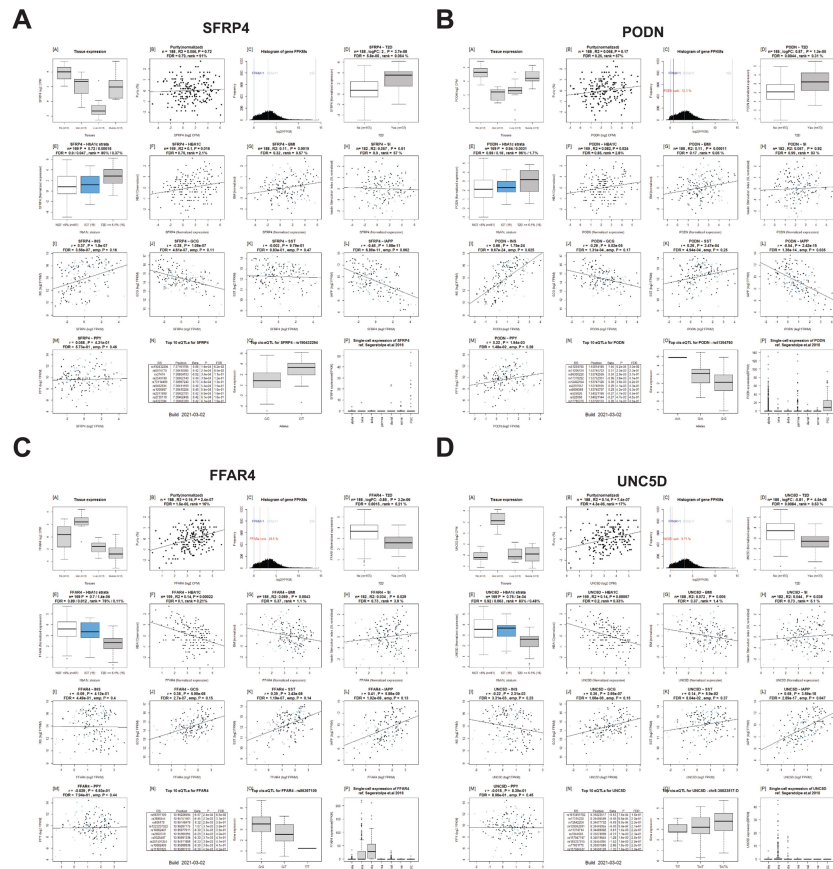


Figure S1.

Islet Gene View results of DEGs found in both Marselli et al. and Asplund et al.

SFRP4 (A), *PODN* (B), *FFAR4* (C), and *UNC5D* (D) were each significantly regulated in T2D vs. non-T2D human islet RNA-seq data in Marselli et al. and Asplund et al. Plots shown were generated using Islet Gene View browser (<https://mae.crc.med.lu.se/IsletGeneView/>). *FFAR4* and *UNC5D* were both down-regulated in T2D and *SFRP4* and *PODN* were both up-regulated.

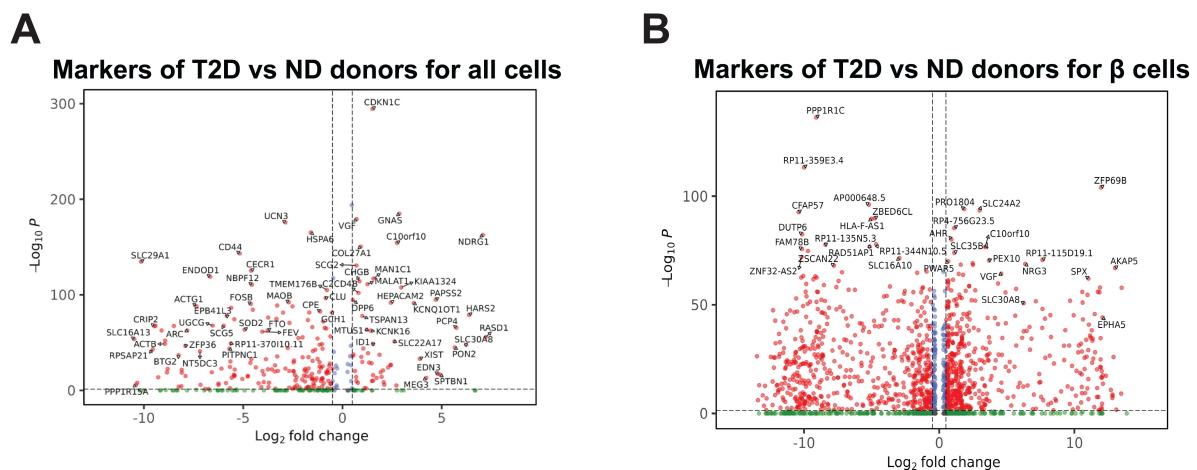


Figure S2.

Human islet single cell RNA-seq clusters and diabetes marker genes for all cells and β -cells.

A) FindClusterMarkers between all single cells from T2D donors vs. non-diabetic (ND) donors. **B)** β -cells reclustered and **(C)** labeled based on donor T2D status. **D)** FindClusterMarkers between β -cells from T2D vs. ND donor single cell data.

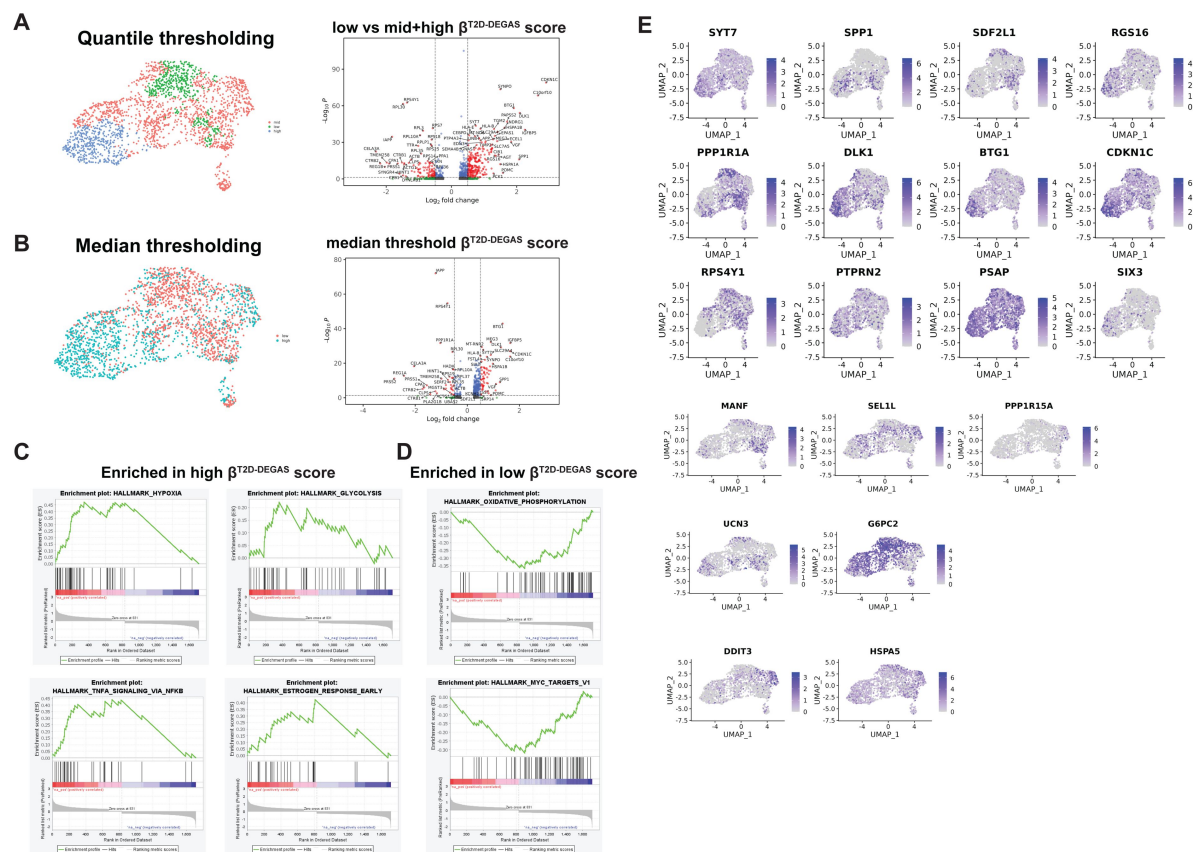


Figure S3.

Thresholding analysis for $\beta^{T2D-DEGAS}$ scores and differential gene expression analysis and plots.

A) Quantile thresholding and volcano plot comparing the upper 20% of $\beta^{T2D-DEGAS}$ scores to the lower 80%. **B)** Median thresholding and volcano plot comparing the upper 50% vs lower 50% of $\beta^{T2D-DEGAS}$ scoring cells. **C)** GSEA results for high $\beta^{T2D-DEGAS}$ scoring cells. **D)** GSEA results of low-scoring $\beta^{T2D-DEGAS}$ scoring cells. **E)** Gene expression overlays on β -cells of selected genes that drive enrichment of GO terms and GSEA categories.

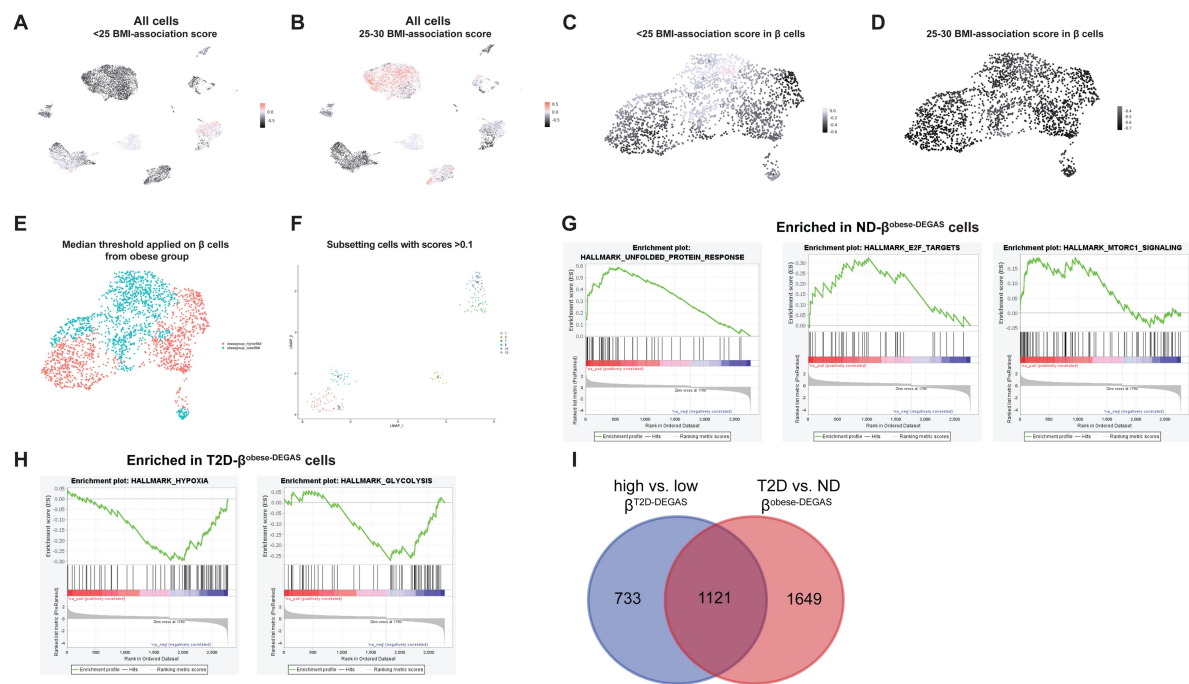


Figure S4.

Score overlay on all islet cells for β lean-DEGAS

(A) and β overweight-DEGAS (B). C, D) Similar score overlays are also shown for β -cells alone. E) Median thresholding of β obese-DEGAS score. F) Subsetting highest-scoring cells based on β obese-DEGAS score. G) GSEA results for ND- β obese-DEGAS cells. H) GSEA results for T2D- β obese-DEGAS cells. I) Venn diagram showing number of genes in β T2D-DEGAS and β obese-DEGAS.

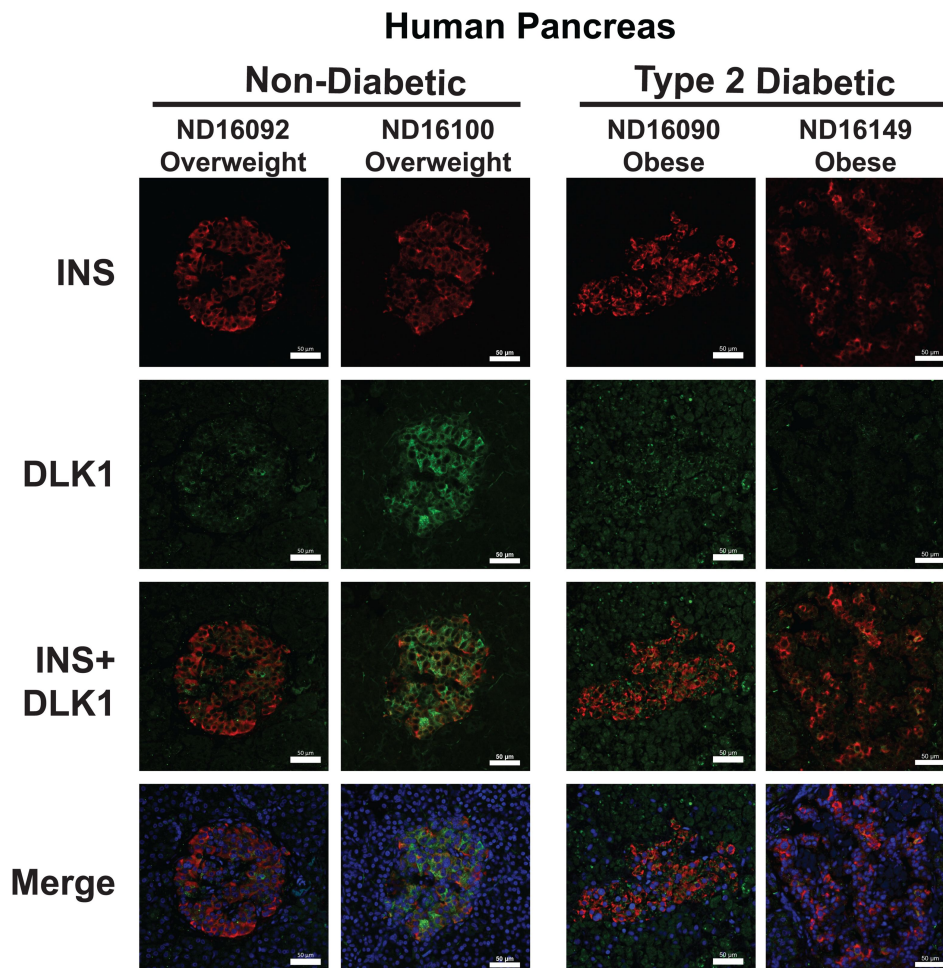


Figure S5.

Formalin-fixed paraffin-embedded (FFPE) sections of human pancreas from non-diabetic (ND) and T2D donors were stained for INS and DLK1.

Representative images are shown from two different ND and T2D donors.

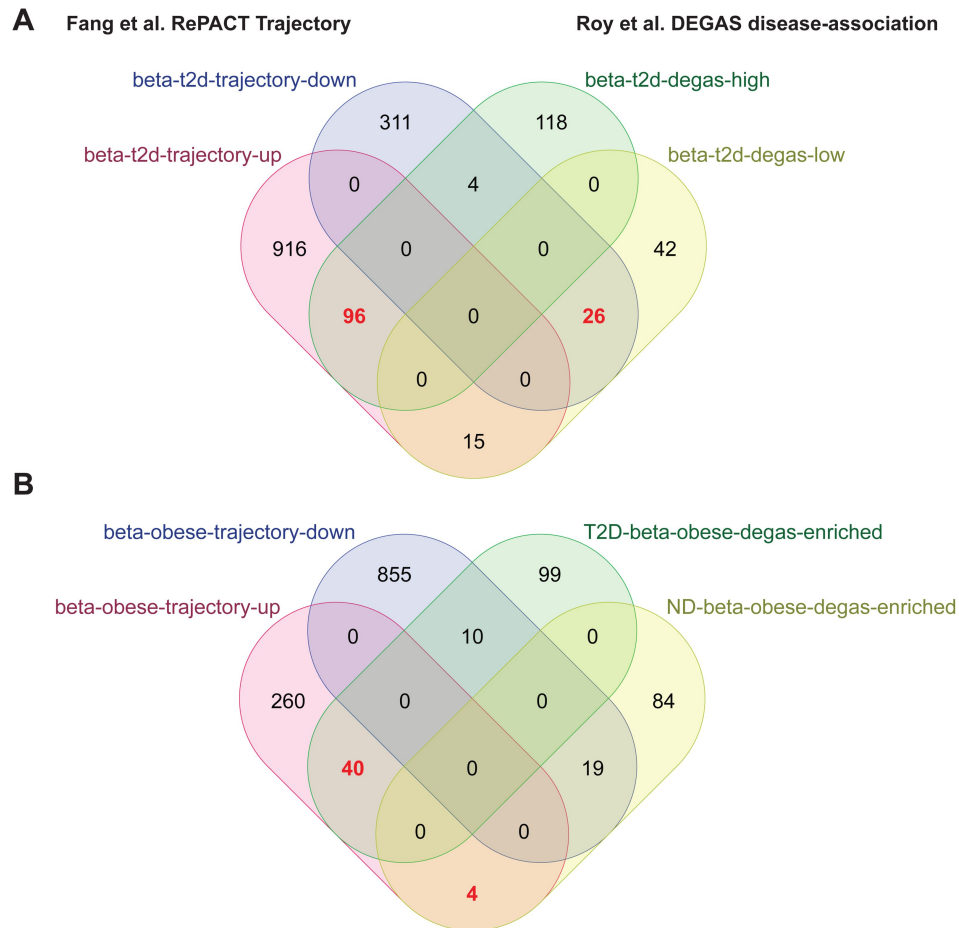


Figure S6.

Overlap of RePACT trajectory genes with genes identified by DEGAS in human islet scRNA-seq data.

Genes from β T2D-DEGAS (**A**), or ND- β obese-DEGAS and T2D- β obese-DEGAS (**B**) that had adjusted p-values < 0.05 and $|\log_2\text{fold-changed}| > 0.58$ were selected for comparison to Fang et al. Table S5 T2D and obesity trajectory genes, respectively.

Data availability

The raw data sets supporting the results of this article are available in the NCBI GEO and the ArrayExpress repositories under the following persistent identifiers (also shown in Table S1): bulk RNA-seq: GSE159984; scRNA-seq: GSE84133, GSE85241, GSE86469, E-MTAB-5061, GSE81608.

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Additional information

Consent for publication

Human islet transcriptomic data used in this project are all published datasets and the donors are deidentified. Human pancreatic tissue sections were obtained from NDRI and were deidentified. Studies on deidentified human tissue or publicly available human transcriptomics datasets do not constitute human research as defined in 45 CFR 46.102(f).

Availability of source code and requirements

- Project name: Islet DEGAS
- Project home page: https://github.com/kalwatlab/Islet_DEGAS_v1 
- Operating system(s): Platform independent
- Programming language: R and Python
- Other requirements: not applicable
- License: ©

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Authors' contributions

G.R., R.S., O.L., S.R., S.D.M., A.M.M., T.S.J., and M.A.K. performed data analyses. G.R. and D.R. performed experiments. M.A.K. and T.S.J. conceived and initiated this project. M.A.K. and T.S.J. supervised the project.

A.M.M. provided processed data files. G.R., T.S.J. and M.A.K. wrote the manuscript. All authors read and approved the manuscript.

List of abbreviations

- ND: non-diabetic
- T1D: type 1 diabetes
- T2D: type 2 diabetes
- RNA-seq: RNA sequencing
- scRNA-seq: single cell RNA sequencing
- FFPE: formalin-fixed paraffin-embedded
- DEGAS: Diagnostic Evidence GAuge of Single cells
- DLK1: Delta Like Non-Canonical Notch Ligand 1
- MANE: Matched Annotation from NCBI and EBI
- RRID: Research Resource identifier
- GSEA: gene set enrichment analysis
- MSigDB: Molecular signatures database
- BMI: body mass index
- UMAP: uniform manifold approximation and projection
- β T2D-DEGAS: T2D-DEGAS disease association score for beta cells
- β obese-DEGAS: obese-DEGAS disease association score for beta cells
- ND- β obese-DEGAS: obese-DEGAS disease association score for beta cells from ND donors
- T2D- β obese-DEGAS: obese-DEGAS disease association score for beta cells from T2D donors
- GO: gene ontology
- RePACT: regressing principle components for the assembly of continuous trajectory
- NDS: normal donkey serum
- PBS: phosphate buffered saline

Additional files

Table S1. List of bulk and single cell transcriptomic datasets. [🔗](#)

Table S2. Bulk RNA-seq metadata, analysis, and comparisons to publicly-available datasets. [🔗](#)

Table S3. Differential gene expression in DEGAS-identified β -cell subpopulations. [🔗](#)

Table S4. Pathway analysis results. [🔗](#)

Table S5. Comparison of DEGAS T2D and obesity genes to Fang et al. RePACT trajectory genes. [🔗](#)

Table S6. FFPE human pancreas metadata. [🔗](#)

Table S7. Antibodies. [🔗](#)

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Reviewer #1 (Public review):

In this manuscript, Roy et al. used the previously published deep transfer learning tool, DEGAS, to map disease associations onto single-cell RNA-seq data from bulk expression data. The authors performed independent runs of DEGAS using T2D or obesity status and identified distinct β -cell subpopulations. β -cells with high obese-DEGAS scores contained two subpopulations derived largely from either non-diabetic or T2D donors. Finally, immunostaining using human pancreas sections from healthy and T2D donors validated the heterogeneous expression and depletion of DLK1 in T2D islets.

Strengths:

- (1) This meta-analysis of previously published scRNA-seq data uses a deep transfer learning tool.
- (2) Identification of novel beta cell subclusters.
- (3) Identified a relatively innovative role of DLK1 in T2D disease progression.

Comments on revisions:

All previous concerns have been addressed.

<https://doi.org/10.7554/eLife.96713.2.sa2>

Reviewer #2 (Public review):

Summary:

The manuscript by Gitanjali Roy et al. applies deep transfer learning (DEGAS) to assign patient-level disease attributes (metadata) to single cells of T2D and non-diabetic patients, including obese patients. This led to the identification of a singular cluster of T2D-associated β -cells; and two subpopulations of obese- β -cells derived from either non-diabetic or T2D donors. The objective was to identify novel and established genes implicated in T2D and obesity. Their final goal is to validate their findings at the protein level using immunohistochemistry of pancreas tissue from non-diabetic and T2D organ donors.

Strengths:

This paper is well-written, and the findings are relevant for β -cell heterogeneity in T2D and obesity.

Weaknesses:

The validation they provide is not sufficiently strong: no DLK1 immunohistochemistry is shown of obese patient-derived sections. Additional presumptive relevant candidates from this transcriptomic analysis should be screened for, at the protein level.

Comments on revisions:

The authors have largely addressed my comments. No further experiments are requested.

<https://doi.org/10.7554/eLife.96713.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

In this manuscript, Roy et al. used the previously published deep transfer learning tool, DEGAS, to map disease associations onto single-cell RNA-seq data from bulk expression data. The authors performed independent runs of DEGAS using T2D or obesity status and identified distinct β -cell subpopulations. β -cells with high obese-DEGAS scores contained two subpopulations derived largely from either non-diabetic or T2D donors. Finally, immunostaining using human pancreas sections from healthy and T2D donors validated the heterogeneous expression and depletion of DLK1 in T2D islets.

Strengths:

- (1) This meta-analysis of previously published scRNA-seq data using a deep transfer learning tool.*
- (2) Identification of novel beta cell subclusters.*
- (3) Identified a relatively innovative role of DLK1 in T2D disease progression.*

Thank you for your comments on the strengths of our work.

Weaknesses :

"There is little overlap of the DE list of bulk RNA-seq analysis in Figure 1D and 1E overlap with the DE list of pseudo-bulk RNA-seq analysis of all cells in Figure S2C. "

Thank you for pointing this out. To clarify, we did not perform pseudo-bulk analysis on the scRNAseq data. Instead, we used the Seurat FindClusterMarkers function to identify differentially enriched genes between T2D and ND single cells. Indeed, there are many significant genes in new Fig S2D (original S2C). There is some overlap between those data and the DEGs from bulk RNAseq data in Fig 1D, including IAPP, ENTPD3, and FFAR4. However, the limited overlap supports the notion that improved approaches are necessary to identify candidate DEGs from single cell data, as simply performing a comparison of T2D to ND of all β -cells may miss important genes or include many false positives. We have now added clarification to the text to highlight this point.

The biological meaning of "beta cells had the lowest scores compared to other cell types" is not clear.

The relatively lower T2D-DEGAS scores for beta cells overall compared to all other cell types (alpha cells, acinar cells, etc) likely reflects the fact that in T2D, beta cell-specific genes can be downregulated. This affects the DEGAS model which is reflected in the scores of all cells in the scRNAseq data. By subsetting the beta cells and replotting them on their own, we can analyze the relative differences in DEGAS scores between different subsets of beta cells. We have now amended the text to clarify, as follows:

“We next mapped the T2D-association scores onto the single cells (Fig 3A). β -cells had a wide distribution of scores, possibly reflecting β -cell heterogeneity or altered β -cell gene expression after onset of T2D (Fig 3B).”

The figures and supplemental figures were not cited following the sequence, which makes the manuscript very difficult to read. Some supplemental figures, such as Figures S1C-S1D, S2B-S2E, S3A-S3B, were not cited or mentioned in the text.

We apologize for this oversight and have now amended the text to call out all figures/panels in order of first introduction.

In Figure 7, the current resolution is too low to determine the localization of DLK1.

We have confirmed that in our Adobe Illustrator file, each microscopy panel has a DPI of >600. We have also provided the highest quality TIFF file versions of our figure set. We hope the reviewer will have access to download the high-quality TIFF file for Fig 7 if possible, or the editorial staff can provide it.

As a result of addressing the critiques, we identified CDKN1C as another promising candidate enriched in the $\beta^{\text{T2D-DEGAS}}$ and $\beta^{\text{obese-DEGAS}}$ subpopulations of β -cells. We found that CDKN1C is heterogeneously expressed at the protein level in β -cells and that it is increased in T2D in agreement with the DEGAS predictions. We have amended the manuscript to highlight CDKN1C more prominently while still discussing DLK1. DLK1 is very interesting, but exhibits greater donor to donor variability in its alterations in T2D.

Reviewer #2 (Public Review):

Summary:

The manuscript by Gitanjali Roy et al. applies deep transfer learning (DEGAS) to assign patient-level disease attributes (metadata) to single cells of T2D and non-diabetic patients, including obese patients. This led to the identification of a singular cluster of T2D-associated β -cells; and two subpopulations of obese- β -cells derived from either non-diabetic or T2D donors. The objective was to identify novel and established genes implicated in T2D and obesity. Their final goal is to validate their findings at the protein

level using immunohistochemistry of pancreas tissue from non-diabetic and T2D organ donors.

Strengths:

This paper is well-written, and the findings are relevant for β -cell heterogeneity in T2D and obesity.

Thank you for your comments on the positive aspects of our work.

Weaknesses:

The validation they provide is not sufficiently strong: no DLK1 immunohistochemistry is shown of obese patient-derived sections.

We have acquired additional FFPE pancreas samples from the Integrated Islet Distribution Program (IIDP) from lean, overweight, and obese humans with and without T2D. We have now stained for CDKN1C and DLK1 in these samples and have integrated the data into Fig 7 and Fig S5.

Because the data with CDKN1C was more striking and consistent with the DEGAS predictions, we have chosen to highlight CDKN1C in the main figure and text. The DLK1 data is still quite interesting, although there is substantial variability between T2D donors when it comes to altered staining intensity. DLK1 presents an interesting challenge, given multiple isoforms and cleavage products, and will require further investigation as the focus of a different manuscript.

Additional presumptive relevant candidates from this transcriptomic analysis should be screened for, at the protein level.

Thank you for this suggestion. We also identified CDKN1C as promising candidate enriched in the $\beta^{\text{T2D-DEGAS}}$ and $\beta^{\text{obese-DEGAS}}$ subpopulations of β -cells. We found that CDKN1C is heterogeneously expressed at the protein level in β -cells and that it is increased in T2D in agreement with the DEGAS predictions. We have amended the manuscript to highlight CDKN1C more prominently while still discussing DLK1. DLK1 is very interesting but exhibits greater donor to donor variability in its alterations in T2D.

Reviewer #1 (Recommendations For The Authors):

Please explain and provide the detailed information on what percentage of the DE list of bulk RNA-seq analysis in Figures 1D and 1E overlap with the DE list of pseudo-bulk RNA-seq analysis of all cells in Figure S2C.

Addressed in response to R1 Comment 1.

Please provide the definition of each cluster of UMAP of the merged human islet scRNA-seq data.

In figure panels 2A-B,D-G and 3A, the clusters are now labeled according to the marker genes described in Fig 2C.

The integrative UMAP needs to be included in the main figure.

We have now moved previous Fig S2A and S2B into the main figures as new Fig 2A-B.

All figures and supplemental figures need to be cited following sequence.

Addressed in response to R1 Comment 3.

In Figure 7, high-resolution images are needed to determine the colocalization of INS and DLK1.

Addressed in response to R1 Comment 4.

Reviewer #2 (Recommendations For The Authors):

Results: 124-128: Fig 1H_The error bars seem high, please include whether the boxplots are SEM or SD. Also, more detail on statistics is missing.

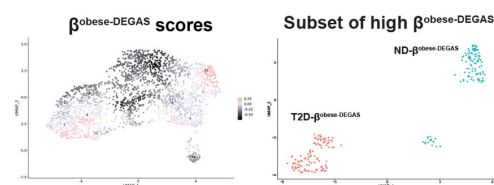
Thank you for pointing out the need for clarification here. The whiskers on the box and whiskers plots are not error bars. By default, in `geom_boxplot()` and `stat_boxplot()`, the whiskers extend to 1.5 times the interquartile range. The box itself represents 50% of the data, the bottom of the box is the first quartile, the middle horizontal line is the median, and the top line of the box is the third quartile. We have now added a clearer description of this to the figure legend and in the methods section.

The genes shown in Fig 1H were selected because they are found in the T2D Knowledge Portal, illustrating a clear link to T2D. At the T2DKP (https://t2d.hugeamp.org/research.html?pageid=mccarthy_t2d_247), PAX4 and APOE are listed as causal, SLC2A2 has strong evidence, and CYTIP has a linked SNP. This is now discussed in the results section before the Fig 1H callout. These genes are significantly differentially expressed using edgeR in panel 1D with FDR<0.05. The individual data points for each human are shown.

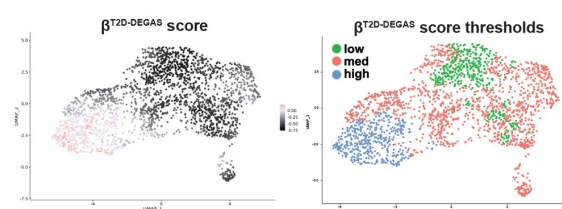
Figure 6: In general, the representation of the data is quite misleading. It would be nice to have an alternative way of presenting the data, especially when comparing beta-obese differentially expressed genes and pathways and T2D beta obese. Maybe an additional Venn diagram can help. Also, it would be nice to compare data from T2D beta nonobese to ND beta obese, especially given how the story is presented in the paper.

Thank you for pointing out this clarity issue. We agree that additional alternate ways to present the data would be helpful. When we performed DEGAS using BMI as the disease feature we noted two major and one minor clusters of high-scoring cells in Fig 6A .

Author response image 1.



Author response image 2.



This contrasted with the score map when we ran DEGAS with T2D as the disease feature

The main difference seems to be the low scoring $\beta^{\text{T2D-DEGAS}}$ cluster is different from the low $\beta^{\text{obese-DEGAS}}$ cluster.

Therefore, we could not easily apply thresholding to the $\beta^{\text{obese-DEGAS}}$ scores, so instead we subsetted them for comparison. It was also apparent from the metadata that single cells from the left-hand side of the β -cell cluster came from donors that had T2D.

To clarify these points and address the reviewer's concerns, we have added a comparison of the DEGs identified for $\beta^{\text{T2D-DEGAS}}$ high vs. low and T2D- $\beta^{\text{obese-DEGAS}}$ vs ND- $\beta^{\text{obese-DEGAS}}$ in Fig S4J, also shown below. DLK1 and CDKN1C fall within the intersection, in addition to being two of the most enriched candidates in each DEGAS run (Fig 4C and Fig 6D).

220-222: Figure 7C_ Is one of the nondiabetic beta samples obese? If so, please clearly label it; if not, that info is missing. One would expect that the DLK1 expression in ND obese beta cells resembles the T2D beta cell and not ND non-obese beta cells. That's a big point of this entire work, and experimentally missing. Additional candidate proteins should be checked.

We have amended the entire Fig 7 to include more data for DLK1 staining as well as adding staining for CDKN1C. We also used CellProfiler to quantify the intensity distribution of DLK1 staining in β -cells and overall found that our initial conclusions were not supported when considering an increased sample size. DLK1 expression is heterogeneous both within and between donors. While we have data from T2D donors that shows DLK1 is lost, other T2D samples indicate that DLK1 is not always lost. At least in the current sample set we have analyzed, we cannot conclude that there is a clear correlation between diabetes or BMI for DLK1. Why DLK1 labels some β -cells and not others and what the role of this subpopulation is an open question.

Alternatively, we greatly appreciate the reviewer's suggestion to validate additional candidates, as this led us to CDKN1C. In new Fig 7E-H we now show that CDKN1C is increased in T2D β -cells, in agreement with the DEGAS predictions.

This work shows that machine learning approaches are powerful for identifying potential candidates, but it also highlights the need for these predictions to be validated at the protein level in human samples.

Discussion: Based on lack of supporting IHC data, this is an overstatement:

"DLK1 expression highly overlapped with high scoring $\beta^{\text{T2D-DEGAS}}$ cells (Figure 7A) and with T2D $\beta^{\text{obese-DEGAS}}$ cells (Figure 7B). DLK1 immunostaining primarily colocalized with β -cells in non-diabetic human pancreas (Figure 7C). DLK1 showed heterogeneous expression within islets and between islets within the same pancreas section, wherein some islets had DLK1/INS co-staining in most β -cells and other islets had only a few DLK1+ β -cells. In the T2D pancreas, DLK1 staining was much less intense and in fewer β -cells, yet DLK1+/INS+ cells were observed (Figure 7C). This contrasts with the relatively higher DLK1 gene expression seen in the β -cells from the $\beta^{\text{T2D-DEGAS}}$ and T2D- $\beta^{\text{obese-DEGAS}}$ subpopulations (Figure 4D & 6C) as highlighted in Figure 7A,B. which were up- or down-regulated in subpopulations of β -cells identified by DEGAS, and to validate our findings at the protein level using immunohistochemistry of pancreas tissue from non-diabetic and T2D organ donors."

This part was at the very end of the last results subsection. This section has been largely rewritten to better describe the new figure and the language has been tempered to not

overinterpret the data shown.

“Our current findings applying DEGAS to islet data have implications for β -cell heterogeneity in T2D and obesity. The abundance of T2D-related factors and functional β -cell genes in our analysis validates applying DEGAS to islet data to identify disease-associated phenotypes and increase confidence in the novel candidate.”

This part was found at the end of the Background section. We have removed the second sentence to temper the language.

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